

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

M.Sc. Digital Forensics and Information Security

Semester – I – December - 2025

Subject Code: CTMSDFIS S1 P4

Date: 19/12/2025

Subject Name: Python and Scripting

Time: 2.00 PM to 5.00 PM

Total Marks: 100

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1	Attempt any three.	
(a)	Explain the architecture of a Linux system with special reference to the role of the Kernel and Shell.	08
(b)	Write and explain the commonly used file manipulation commands (<code>cp</code> , <code>mv</code> , <code>rm</code> , <code>mkdir</code> , <code>ln</code>).	08
(c)	Explain absolute and relative pathnames in Linux. Provide suitable examples and demonstrate how <code>cd</code> , <code>pwd</code> , and <code>ls</code> commands help in navigation.	08
(d)	Explain the different network configuration commands in Linux. Discuss <code>ifconfig</code> , <code>ip</code> , <code>ping</code> , <code>netstat</code> , and <code>traceroute</code> with examples.	08
Q.2	Attempt any three.	
(a)	Explain the different types of loops in shell scripting with suitable examples for each.	08
(b)	Explain branching (decision-making) in shell scripting with suitable examples.	08
(c)	Write a shell script to check whether a given number is even or odd.	08
(d)	Explain Lists, Dictionaries, and Sets in Python. Compare their properties, operations, and typical use cases with suitable examples.	08
Q.3	Attempt any three.	
(a)	Write a Python program to calculate the factorial of a number using a loop.	08
(b)	Write a program in Python to take a prompt and take a string as input from user then count the number of vowels in the given string.	08
(c)	Explain the concepts of classes and objects in Python with an example.	08
(d)	Explain the concept of functions in Python. Discuss user-defined functions, function definitions, return statements, and parameter types with suitable examples.	08
Q.4	Attempt any two.	
(a)	Write a Python program that reads a text file line by line, handles possible exceptions and prints the total number of lines in the file. Explain how exception handling ensures the program does not crash.	07

- (b) Explain the role of Sys Module & OS Module with examples. 07
- (c) Explain the use of the requests module using any five commonly used functions provided by the module. 07

Q.5

Attempt any two.

- (a) What types of operators are there in PowerShell? Explain with examples. 07
- (b) Explain the concept of a PowerShell cmdlet and illustrate your answer with four examples of commonly used cmdlets. 07
- (c) Write a PowerShell script to print numbers from 1 to 10 using a loop. 07

END OF PAPER

#

$n = \text{input}('')$

fact = 1

$i = 1$

while

$i \leq n$:

fact *= i

$i += 1$

print(fact)

eq
gt
lt
le
ge

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

$$1 \times 1 = 1$$

$$1 + 1 = 2$$

$$2 < 5$$

$$1 \times 2 = 2$$

$$3$$

$$6$$

$$24 \times 5$$

$$= 120$$